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Advances in Space Research 78 (2026) 5267–5278

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

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Multi-threaded CPU implementation of the Holmes algorithm for pointwise high-degree gravitational potential and acceleration evaluation

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Received 5 March 2026; received in revised form 22 June 2026; accepted 24 June 2026
Available online 26 June 2026

Abstract

Accurate and efficient evaluation of Earth's gravitational potential using high-degree spherical harmonic expansions is essential for precise long-term orbit propagation in celestial mechanics. Widely used recursive algorithms, such as those developed by Cunningham, Belikov, and Holmes, become computationally demanding at ultra-high degrees required by modern global gravity models. This paper introduces a multi-threaded parallel implementation of Holmes' algorithm, designed for evaluations on standard multi-core CPUs. The method was extensively tested on various processor architectures and across a range of maximum harmonic degrees up to 2000. By distributing the outer order-wise summation across independent threads, the approach gives speedups of up to 8.4 times. Numerical tests confirm that the parallel version fully preserves the accuracy and stability of the serial Holmes algorithm. All algorithms discussed, including the parallel implementation, are freely available on GitHub.
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2010 MSC: 85–04

1. Introduction

Accurate orbit modeling is essential for the success of any space mission. To ensure the closest possible agreement between predicted and actual satellite trajectories, the force model must be as complete and precise as possible. For near-Earth satellites, the dominant force is Earth's gravitational attraction.

Earth's gravitational field is highly inhomogeneous. Before the space era, knowledge of the field came entirely from surface measurements, which offered high resolution in limited areas but no global coverage. The advent of artificial satellites made global modeling possible, since field irregularities act as significant perturbations in orbital motion. Missions such as CHAMP (Reigber et al., 2003), GRACE (Tapley et al., 2004) (later continued as GRACE-FO (Kornfeld et al., 2019)), and GOCE (Johannessen et al., 2003) have dramatically improved the accuracy and resolution of these global models.

Mathematical approaches for determining the gravitational field include spherical harmonic expansions (Heiskanen and Moritz, 1967; Pines, 1973; Cunningham, 1970; Belikov and Tajbatorov, 1990; Fukushima, 2011; Holmes and Featherstone, 2002; Chen et al., 2019), spheroidal harmonics expansions (Thong and Grafarend, 1989;

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https://doi.org/10.1016/j.asr.2026.06.089
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Reimond and Baur, 2016), Fourier-domain techniques (Shin et al., 2006; Pedersen et al., 2015; Wu, 2018), and partial differential equation-based approaches (Farquharson and Mosher, 2009; Jahandari and Farquharson, 2013; Long and Farquharson, 2019). Spherical harmonic expansion, the most widely used method, expresses the potential as a weighted sum of spherical harmonics with coefficients known as Stokes coefficients. It forms the basis of most orbit propagation software, including GMAT (Hughes, 2007). Although the series is infinite in theory, practical implementations truncate it at a finite degree and order, which determines the model's resolution. For example, EGM96 (Lemoine et al., 1998) provides coefficients up to degree and order 360, while EGM2008 (Pavlis et al., 2012) extends to degree 2190 and order 2159. Well-known algorithms for evaluating such expansions include those proposed by Cunningham (Cunningham, 1970), Belikov (Belikov and Tajbatorov, 1990), and Holmes (Holmes and Featherstone, 2002), which are described in detail later in the paper.

The efficiency of numerical orbit integrators depends critically on the speed of evaluating the accelerations in the equations of motion (Montenbruck and Gill, 2000). Since gravitational acceleration must be computed at every integration step, the performance of the gravity field evaluation routine is a key factor. Parallel computing offers one of the most effective ways to accelerate such calculations. Packages originally designed for cosmic microwave background analysis – including HealPix (https://irsa.ipac.caltech.edu/healpix/), GLESP (https://www.glesp.nbi.dk/), LIBPSHT (Reinecke, 2010), and S^2HAT (Stompor, 2011) – allow for synthesis of continuous fields. In celestial mechanics, however, fast evaluation of spherical harmonic sums is vital for long-term orbit propagation over intervals that may span years. GPU-based parallel implementations have been investigated in several studies (Hupca et al., 2012; Belkner et al., 2024; Cheng et al., 2025; Rubio et al., 2024). Their main limitation is the small shared memory capacity, which restricts efficient synthesis and makes them better suited to analysis tasks (Hupca et al., 2012).

Most multi-threaded CPU implementations as in works (Xiao and Lu, 2007; Goli and Foroughi, 2025; Szydlarski et al., 2011) provide computational acceleration only for regional models like EGMWP (Lee et al., 2013). However, orbital propagation requires continuous evaluation of the gravitational field at a spacecraft's precise and changing location, not a one-time calculation on a predefined grid.

Celestial mechanics and astrodynamics applications require repeated pointwise evaluations at arbitrary positions and altitudes. While some advances exist in adaptive truncation, parallel implementations, and specialized recursions, faster alternatives to the classical recursive algorithms (such as those of Cunningham, Holmes, Fukushima, and Belikov) have not been widely adopted for long-term propagation on standard hardware.

In this paper, we present a multi-threaded parallel implementation of Holmes' algorithm (Holmes and Featherstone, 2002) for computing the Earth's gravitational potential for pointwise evaluation problems. We evaluate its performance over a set of initial conditions as well as its accuracy compared to the methods of Cunningham (Cunningham, 1970) and Belikov (Belikov and Tajbatorov, 1990). The approach is particularly suitable for use on standard personal computers with a modest number of cores.

The rest of the paper is structured as follows. Section 2 reviews the standard algorithms of Cunningham, Belikov, and Holmes. Section 3 describes the parallelization strategy applied to Holmes' method. Section 4 presents comparative results for accuracy and computational speed. Finally, the paper concludes with final remarks.

2. Gravitational potential computation algorithms

The motion of an artificial Earth satellite cannot be adequately described using purely Keplerian orbital elements, as the Earth deviates significantly from a point mass with uniform density. Mass inhomogeneities within the Earth produce a nonuniform gravitational field, which perturbs the satellite's trajectory away from the ideal two-body Keplerian motion. The Earth's gravitational potential outside the planet is conventionally represented by an expansion in spherical harmonics:

$$V = \frac{GM_\oplus}{r}\sum_{n=0}^{\infty}\sum_{m=0}^{n}\frac{a^n}{r^n}P_{nm}(\cos(\theta))(C_{nm}\cos(m\lambda)+S_{nm}\sin(m\lambda)), \tag{1}$$

where (r,θ,λ) – point spherical coordinates ($0 \leqslant \theta \leqslant \pi$, $0 \leqslant \lambda \leqslant 2\pi$), a – equatorial Earth radius, $GM_\oplus$ – gravitational parameter of the Earth, P_{nm} – associated Legendre function (ALF) of degree n and order m.

However, this expansion poses a computational challenge, as the expansion coefficients C_{nm} and S_{nm} differ by tens of orders of magnitude even at low expansion degrees. Therefore, global models such as EGM96, EGM2008, etc. use normalized coefficients, which have much closer values:

$$\overline{C}_{nm}, \overline{S}_{nm} \approx \frac{10^{-5}}{n^2} \tag{2}$$

These coefficients, derived primarily from satellite observations, are usually tabulated. In this section we review the three most widely used approaches to solving the synthesis problem.

2.1. Cunningham's method

One of the first algorithms offering efficient computation of the Earth's gravitational field using spherical harmonics was proposed by Cunningham in 1969 (Cunningham, 1970). This method introduces quantities $V_{n,m}$ such that the gravitational potential can be represented as:

$$V = \mathrm{Re}\sum_{n=0}^{\infty}\sum_{m=0}^{n} a^n (C_{nm} - iS_{nm}) V_{n,m}, \tag{3}$$

where i – imaginary unit, Re – operation of taking real part of the complex number.

To calculate the values of $V_{n,m}$, it is proposed to use recurrent relations:

$$\begin{aligned} V_{0,0} &= \tfrac{1}{r}, \\ V_{n,n} &= (2n-1)\tfrac{x+iy}{r^2} V_{n-1,n-1}, \\ (n-m)V_{n,m} &= (2n-1)\tfrac{z}{r^2} V_{n-1,m} - \tfrac{n+m-1}{r^2} V_{n-2,m}, \end{aligned} \tag{4}$$

where x, y, z – cartesian coordinates of a point in Earth-fixed frame, r – geocentric distance of a point.

Although the original paper employs complex-valued quantities, it leaves practical implementation details to the reader. For practical implementation, the book by Montenbruck and Gill (Montenbruck and Gill, 2000, pp. 66–68) provides useful guidance. It suggests splitting one value with complex values into two separate values V_{nm} and W_{nm} with recursion relations:

$$\begin{aligned} V_{mm} &= (2m-1)\left(\tfrac{x\cdot a}{r^2} V_{m-1,m-1} - \tfrac{y\cdot a}{r^2} W_{m-1,m-1}\right), \\ W_{mm} &= (2m-1)\left(\tfrac{x\cdot a}{r^2} W_{m-1,m-1} + \tfrac{y\cdot a}{r^2} V_{m-1,m-1}\right) \end{aligned} \tag{5}$$

and

$$\begin{aligned} V_{nm} = \left(\tfrac{2n-1}{n-m}\right)\cdot\tfrac{z\cdot a}{r^2} \cdot V_{n-1,m} - \\ - \left(\tfrac{n+m-1}{n-m}\right)\cdot\tfrac{a^2}{r^2}\cdot V_{n-2,m}, \\ W_{nm} = \left(\tfrac{2n-1}{n-m}\right)\cdot\tfrac{z\cdot a}{r^2} \cdot W_{n-1,m} - \\ - \left(\tfrac{n+m-1}{n-m}\right)\tfrac{a^2}{r^2}\cdot W_{n-2,m} \end{aligned} \tag{6}$$

with the initial values $V_{00} = \frac{a}{r}$ and $W_{00} = 0$. To obtain all required values V_{nm} and W_{nm} the recursion can be computed following the scheme shown in Fig. 1.

A sum over n and m of terms which linearly depend on the quantities V_{nm} and W_{nm} expresses components of the perturbing accelerations:

$$\begin{aligned} \ddot{x}_{nm} &\overset{(m=0)}{=} \tfrac{GM_\oplus}{a^2}(-C_{n0}V_{n+1,1}), \\ \ddot{x}_{nm} &\overset{(m>0)}{=} \tfrac{GM_\oplus}{2a^2}\left[(-C_{nm}V_{n+1,m+1} - S_{nm}W_{n+1,m+1}) + \right. \\ &\left. + \tfrac{(n-m+2)!}{(n-m)!}(C_{nm}V_{n+1,m-1} + S_{nm}W_{n+1,m-1})\right]. \end{aligned} \tag{7}$$

Relations for the y- and z-components are different only in order of V_{nm} and W_{nm} and can be found in the book (Montenbruck and Gill, 2000, Eq. 3.33).

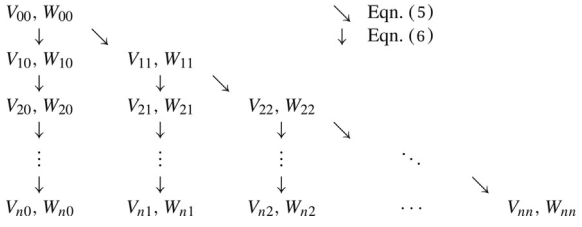


Fig. 1. Recursion scheme in Cunningham's algorithm.

2.2. Belikov's method

Unlike Cunningham's method, Belikov and Taibatorov's method (Belikov and Tajbatorov, 1990) avoids complex numbers and instead operates directly in spherical coordinates (see Eq. 1).

This method utilizes column recursion for ALFs with special normalization which result in recurrent relations:

$$\begin{aligned} \widehat{P}_{nm}(\cos(\theta)) &= \cos(\theta)\widehat{P}_{n-1,m}(\cos(\theta)) - \\ &- \tfrac{\sin(\theta)}{4}\left[\widehat{P}_{n-1,m+1}(\cos(\theta)) - 4\widehat{P}_{n-1,m-1}(\cos(\theta))\right]. \end{aligned} \tag{8}$$

The summation order changes to:

$$\begin{aligned} V(r,\theta,\lambda) &= \tfrac{GM_\oplus}{r} + \\ &+ \tfrac{GM_\oplus}{r}\sum_{m=0}^{M}\left[\cos(m\lambda)\sum_{n=\mu}^{N}\left(\tfrac{a}{r}\right)^n \widehat{C}_{nm}\widehat{P}_{nm}(\cos(\theta)) + \right. \\ &\left. + \sin(m\lambda)\sum_{n=\mu}^{N}\left(\tfrac{a}{r}\right)^n \widehat{S}_{nm}\widehat{P}_{nm}(\cos(\theta))\right], \end{aligned} \tag{9}$$

where $\mu = \max(2, m)$, N, M – maximum degree and order of the expansion.

Such normalization of the ALFs requires the corresponding normalization of the Stokes coefficients:

$$\begin{pmatrix} \widehat{C}_{nm} \\ \widehat{S}_{nm} \end{pmatrix} = \frac{\sqrt{2(2n+1)(n+m)!(n-m)!}}{2^m n!}\begin{pmatrix} \overline{C}_{nm} \\ \overline{S}_{nm} \end{pmatrix}. \tag{10}$$

The column recursion algorithm, illustrated in Fig. 2, derives its name from the fact that it uses ALFs of the same degree (or order) on the right-hand side of the equation.

For computational simplicity the authors use function T_{nm}:

$$T_{nm} = \frac{a^n}{r^n}\widehat{P}_{nm}(\cos(\theta)), \tag{11}$$

which also follows a column recursion.

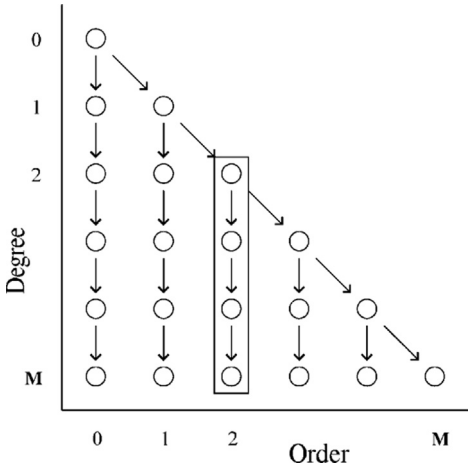


Fig. 2. Column recursion scheme for a constant order algorithm.

Belikov and Taibatorov describe gravitational potential in spherical coordinates which leads to a well-known singularity in polar regions while calculating perturbing accelerations. This singularity can be easily avoided by introducing yet another auxiliary function:

$$D_{nm} = T_{nm}/\sin(\theta), n \geqslant 0, 1 \leqslant m \leqslant n. \tag{12}$$

This function D_{nm} also follows column recursion and doesn't have singularities in pole regions. Another way to avoid these singularities was proposed by Petrovskaya and Vershkov (2006).

Calculation of the resulting perturbing accelerations requires three partial derivatives: $\frac{\partial V}{\partial r}, \frac{\partial V}{\partial \theta}, \frac{\partial V}{\partial \lambda}$. All of them can be described as a linear sum of functions T_{nm}, D_{nm} and Stokes coefficients. The article Belikov and Tajbatorov (1990) provides all equations as well as the recursion relations for T_{nm} and D_{nm}.

Eventually, components of perturbing acceleration are represented as a linear combination of partial derivatives of gravitational potential:

$$\begin{aligned} \ddot{x} &= \sin(\theta)\cos(\lambda)\frac{\partial V}{\partial r} + \frac{\cos(\theta)\cos(\lambda)}{r}\frac{\partial V}{\partial \theta} - \frac{\sin(\lambda)}{r\sin(\theta)}\frac{\partial V}{\partial \lambda}; \\ \ddot{y} &= \sin(\theta)\sin(\lambda)\frac{\partial V}{\partial r} + \frac{\cos(\theta)\sin(\lambda)}{r}\frac{\partial V}{\partial \theta} + \frac{\cos(\lambda)}{r\sin(\theta)}\frac{\partial V}{\partial \lambda}; \\ \ddot{z} &= \cos(\theta)\frac{\partial V}{\partial r} - \frac{\sin(\theta)}{r}\frac{\partial V}{\partial \theta}. \end{aligned} \tag{13}$$

Although the method employs spherical coordinates, the coefficients T_{nm} and D_{nm} enable it to model any type of satellite orbit, including polar orbits.

2.3. Holmes's method

The Holmes algorithm (Holmes and Featherstone, 2002) uses ALFs and, accordingly, the coefficients $\overline{C}_{nm}, \overline{S}_{nm}$ introduced in Eq. 2. In this case, as in the Belikov and Taibatorov's method, a column method (see Fig. 2) for calculating the ALFs is used, so the gravitational potential takes form as in Eq. 9.

A distinctive feature of the Holmes' method is the recurrent calculation of the ALFs using relations:

$$\frac{\overline{P}_{nm}}{\overline{P}_{mm}} = a_{nm} t \frac{\overline{P}_{n-1,m}}{\overline{P}_{mm}} - b_{nm}\frac{\overline{P}_{n-2,m}}{\overline{P}_{mm}}, \forall n > m, \tag{14}$$

where $t \equiv \cos\theta$,

$$\begin{aligned} a_{nm} &= \sqrt{\frac{(2n-1)(2n+1)}{(n-m)(n+m)}}, \\ b_{nm} &= \sqrt{\frac{(2n+1)(n+m-1)(n-m-1)}{(n-m)(n+m)(2n-3)}}. \end{aligned} \tag{15}$$

The recursion is initialized with functions with $n = m$ which have a normalized value of one. Their real values are calculated further in the algorithm:

$$\begin{aligned} \overline{P}_{mm} &= \prod_{i=1}^{m} U_i, \\ U_i &= \sin(\theta)\left(\sqrt{3}\delta_{i1} + \sqrt{\tfrac{2i+1}{2i}}(1-\delta_{i1})\right) \ \forall m \geqslant 1, \end{aligned} \tag{16}$$

where δ_{i1} is the Kronecker delta.

The values of $\cos(m\lambda), \sin(m\lambda)$ can also be obtained recursively:

$$\begin{aligned} \sin(m\lambda) &= \sin(\lambda) \quad \cos((m-1)\lambda) + \\ &\quad + \sin((m-1)\lambda)\cos(\lambda); \\ \cos(m\lambda) &= \cos(\lambda) \quad \cos((m-1)\lambda) - \\ &\quad - \sin((m-1)\lambda)\sin(\lambda). \end{aligned} \tag{17}$$

This approach to computing the ALFs avoids limitations caused by the precision of numbers stored in the computer (such as IEEE double), when working with orders $m > 2700$. In Belikov's method, for example, for $\theta = 1^\circ$ and $M = 2700$, the normalized value $\overline{P}_{nm}(\cos(\theta)) \sim 10^{-4747}$, which significantly exceeds the precision of IEEE double (Holmes and Featherstone, 2002).

3. Parallel method

Parallelizing independent computations across multiple threads and aggregating their results in a master thread significantly accelerates program execution.

Holmes' method can be parallelized by splitting the outer order summation (the m-loop) across K threads. Since our algorithm targets CPU execution, we precompute values common to all threads, such as the terms $(a/r)^n, n = \overline{0,N}$. The flowchart of the algorithm is shown in Fig. 3. The most delicate step of the algorithm is the efficient splitting of the m-loop into partial sums. The key goal is to have roughly the same number of terms in each thread. In the general case of summing up to degree N and order M the total number of terms would be:

$$\Sigma = \frac{M + 2N + 2 + 2NM - M^2}{2}, \tag{18}$$

which, when $M = N$ transforms into the sum of the arithmetic progression $(M+1)(M+2)/2$.

Thus, the optimal number of terms in each sum should be $\Sigma_{opt} = \Sigma/K$, where K is the number of threads. We can then determine a set of boundaries $\{M_1, M_2, \ldots, M_{K-1}\}$ such that the first thread sums terms for $m \in [0; M_1]$, the second thread for $m \in (M_1, M_2]$, and so on, up to the last thread for $m \in (M_{K-1}, M]$.

The whole algorithm consists of five steps:

1. calculation of common values for all threads: $\left(\frac{a}{r}\right)^n \left(n = \overline{0,N}\right), \sin\theta, \cos\theta, \sin\lambda, \cos\lambda$
2. calculation of all values of harmonics.$P_{mm}, m = \overline{0,M}$, as well as the values.$\cos((m_i-1)\lambda), \sin((m_i-1)\lambda)$ to start the recurrent calculation using the Eq. 17.
3. calculation of partial sums for individual threads
4. combining partial sums in one master-thread
5. calculation of acceleration components using Eq. 13.

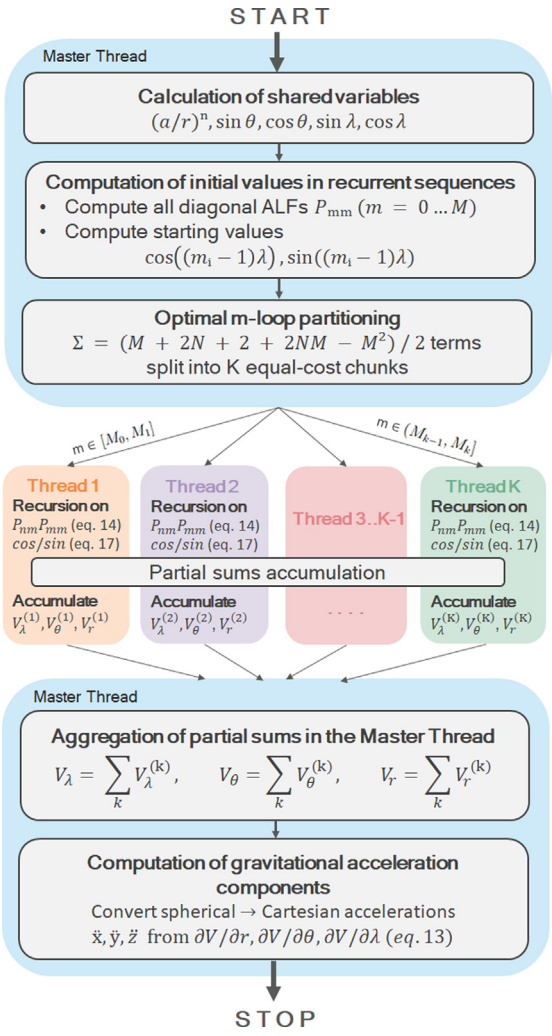


Fig. 3. Multi-threaded Holmes algorithm.

4. Numerical results

In this section, we present numerical tests to compare the performance of parallel Holmes algorithm against classical Cunningham, Belikov and single-threaded Holmes algorithms. There are two different test types that we use. *Random data testing* executes each algorithm several times using randomly generated evaluation points (r, θ, λ) to find its average execution time. This type of test is used for the analyses of speedup, efficiency and execution time. On the other hand, *sequential data testing* examines the accuracy of the computed accelerations. Initial conditions consist of coordinates (r, θ, λ) and velocity with magnitude equal to local circular velocity. Then each algorithm propagates this point according to the computed accelerations. This type of testing is used in the accuracy assessment (Section 4.2). We used the Earth model EGM2008 and equal maximum degree and order of the expansion for all tests.

4.1. Multiple threads speed test

To eliminate the influence of specific processor or operating system characteristics, we tested the algorithm on processors of various families and on Linux and Windows operating systems. The first column of the Table 1 indicates the code name that will be used further to refer to this processor. The "Threshold Degree N" column shows the minimum number of harmonics at which the parallel algorithm is faster than the serial one for each processor.

For each maximum degree in the range from 5 to 2000 with a step 5, the execution time of each algorithm was determined. The execution time was defined as the average execution time of the algorithm over 100 (for degrees from 5 to 360) and 10 (for degrees from 365 to 2000) consecutive runs. The average execution time of the algorithms as a function of the number of harmonic terms and the number of threads for the P6T12 processor (see Table 1) is shown in the Fig. 4. The numbers in the labels indicate the number of threads used. Here we can see that Cunningham's method is nearly an order of magnitude slower than parallel Holmes methods and proves to be the slowest one. Belikov's method is consistently slower than the single-threaded Holmes. Thus, for a maximum degree of 2000 Cunningham, Belikov, and single-threaded Holmes methods have performances of 221, 124 and 100 ms respectively.

It is worth mentioning that Cunningham's method consistently evaluates gravitational potential only up to a maximum degree of expansion of approximately 646. As the degree increases, values of ALFs in forward recursion scheme begin to overflow resulting in NaN (not a number) values of some harmonics, causing the entire sum to become NaN. Both Belikov and Holmes methods hold up to the maximum tested degree of 2000.

The speedup (S_K) and efficiency (E_K) values are commonly used as parameters of a parallel computing algorithm performance:

$$\begin{aligned} S_K &= T_1/T_K, \\ E_K &= S_K/K, \end{aligned} \tag{19}$$

where T_1 – execution time of single-threaded algorithm, T_K – execution time of an algorithm with K threads. The speedup quantifies how much faster the parallel implementation runs compared to the single-threaded version, with ideal scaling yielding $S_K = K$. The efficiency indicates how effectively the additional threads are utilized, where values close to 100% signify near-perfect parallelization and minimal overhead from synchronization or load imbalance. The speedup and efficiency parameters for the different processors, thread counts and maximum harmonic number, are presented in Tables 2 and 3. Figs. 5 and 6 show speedup and efficiency graphs depending on the maximum harmonic count and the number of threads used for the P6T12 processor. The spike-like irregularities

Table 1
Benchmarked processors and parallel algorithm efficiency threshold. The threshold degree N indicates the lowest degree at which parallel computation outperforms the sequential baseline.

Label	Processor specifications	Threshold degree N
P4T8	Intel Core i7-7820HQ (Laptop, Linux)	100
P6T12	AMD Ryzen 5 7600X (Desktop, Windows)	60
P8T16	AMD Ryzen 7 PRO 3700 (Desktop, Windows)	60
P14T20	Intel Core i7-1280P (Laptop, Windows)	100

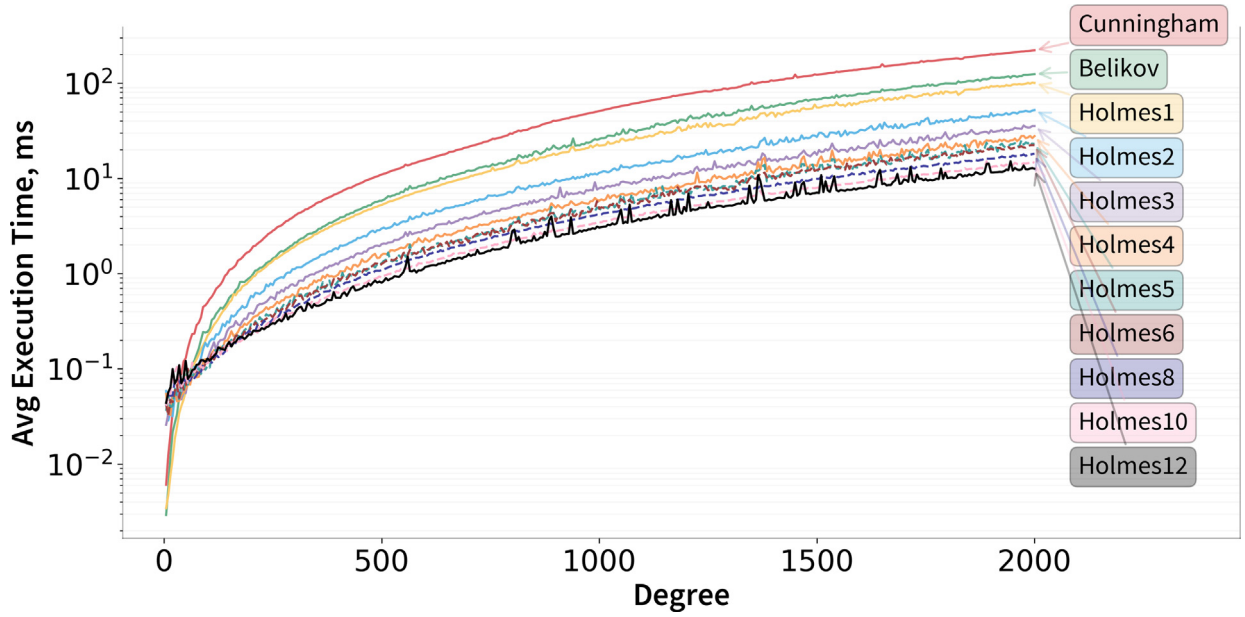


Fig. 4. Average time of each algorithm with respect to maximum degree for the P6T12 processor. Digits on labels indicate number of used threads from 1 to 12.

Table 2
Speedup for Holmes algorithms across different processors and maximum degree.

		Threads													
		2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**14**	**16**	**18**	**20**
n = 100	**P4T8**	1.18	1.61	1.20	1.07	1.09	1.00	0.92	–	–	–	–	–	–	–
	P6T12	1.64	2.12	2.49	2.53	2.35	2.54	2.49	2.51	2.34	2.39	–	–	–	v
	P8T16	1.61	1.87	2.12	2.19	2.12	2.16	2.06	1.95	1.81	1.72	1.47	1.32	–	–
	P14T20	1.40	1.40	1.44	1.63	1.68	1.76	1.71	1.85	2.03	1.88	1.76	1.72	1.54	1.47
n = 360	**P4T8**	1.64	2.27	2.43	2.50	2.77	3.04	3.12	–	–	–	–	–	–	–
	P6T12	1.91	2.76	3.58	3.32	3.95	4.35	4.83	5.20	5.65	5.02	–	–	–	–
	P8T16	1.87	2.71	3.41	4.22	4.89	5.40	4.87	4.84	4.95	5.54	5.98	6.29	–	–
	P14T20	1.75	2.40	3.04	3.41	4.04	3.94	3.94	3.85	3.65	3.60	3.65	3.79	3.97	4.28
n = 720	**P4T8**	1.80	2.54	3.11	2.81	3.01	3.33	3.84	–	–	–	–	–	–	–
	P6T12	1.89	2.72	3.52	4.39	4.18	4.51	4.95	5.50	6.03	6.20	–	–	–	–
	P8T16	1.93	2.82	3.60	4.42	5.06	5.76	5.59	5.30	5.50	6.48	7.44	8.20	–	–
	P14T20	1.82	2.69	3.21	3.70	3.26	3.58	3.85	4.31	4.80	4.90	5.23	5.51	5.53	5.31
n = 2000	**P4T8**	1.83	2.63	3.49	2.48	3.22	3.75	4.31	–	–	–	–	–	–	–
	P6T12	1.92	2.83	3.61	4.32	4.48	5.08	5.53	6.18	6.88	7.93	–	–	–	–
	P8T16	1.92	2.80	3.61	4.40	5.16	6.01	5.09	5.10	5.49	6.45	7.43	8.37	–	–
	P14T20	1.84	2.59	3.02	3.87	4.96	4.03	5.51	4.74	5.74	5.31	5.23	5.95	5.43	5.64

Table 3
Efficiency (in percent) for Holmes algorithms across different processors and maximum degree.

		Threads													
		2	3	4	5	6	7	8	9	10	12	14	16	18	20
n = 100	**P4T8**	59	53	30	21	18	14	11	–	–	–	–	–	–	–
	P6T12	81	70	62	50	39	36	31	27	23	19	–	–	–	–
	P8T16	80	62	53	43	35	30	25	21	18	14	10	8	–	–
	P14T20	69	46	35	32	27	25	21	20	20	15	12	10	8	7
n = 360	**P4T8**	81	75	60	49	46	43	38	–	–	–	–	–	–	–
	P6T12	95	91	89	66	65	62	60	57	56	41	–	–	–	–
	P8T16	93	90	85	84	81	77	60	53	49	46	42	39	–	–
	P14T20	87	79	76	68	67	56	49	42	36	30	26	23	22	21
n = 720	**P4T8**	89	84	77	56	50	47	48	–	–	–	–	–	–	–
	P6T12	94	90	88	87	69	64	61	61	60	51	–	–	–	–
	P8T16	96	93	89	88	84	82	69	58	55	54	53	51	–	–
	P14T20	91	89	80	74	54	51	48	47	48	40	37	34	30	26
n = 2000	**P4T8**	91	87	87	49	53	53	53	–	–	–	–	–	–	–
	P6T12	96	94	90	86	74	72	69	68	68	66	–	–	–	–
	P8T16	95	93	90	87	86	85	63	56	54	53	53	52	–	–
	P14T20	92	86	75	77	82	57	68	52	57	44	37	37	30	28

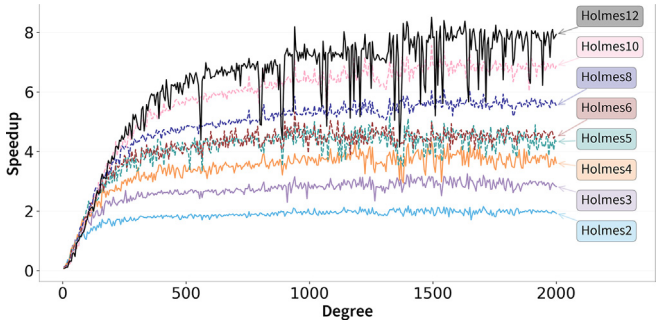


Fig. 5. Speedup of parallel Holmes algorithm over maximum degree for threads from 2 to 12 for processor P6T12.

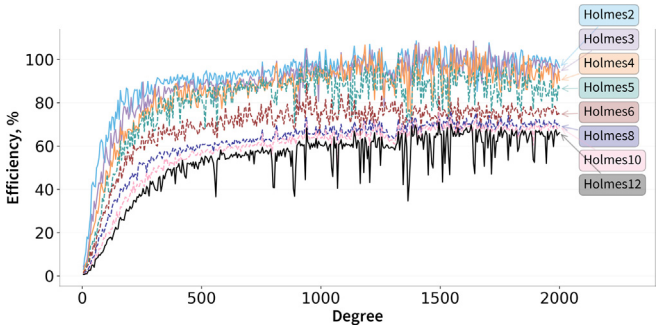


Fig. 6. Efficiency of parallel Holmes algorithm over maximum degree for threads from 2 to 12 for processor P6T12.

in the graphs are caused by background system processes. They become more prominent with increasing number of cores, since parallel execution makes these disturbances more visible.

For a small number of harmonics, standard single-threaded methods can be faster than parallel ones. This behavior is expected (e.g., Nukada and Matsuoka, 2009) and is explained by the fact that splitting the computations into multiple threads, as well as the subsequent merging of the results, are time-consuming procedures. However, for high-order expansions of the gravitational potential ($n \geqslant 100$), the parallel algorithm has $S_K > 1$. Moreover, for the same number of threads, the time gain increases with increasing the order. A similar constant increase is not observed with increasing the number of threads for a fixed number of harmonics. Table 2 shows that, for example, for the P4T8 processor with $n = 720$and 2000, there is a performance drop when moving from 4 to 5 threads. This may be due to the fact that the P4T8 processor has only 4 physical cores, while 8 threads are available thanks to hyper-threading technology. Figs. 5, 6 show similar behavior for the speedup and efficiency values. Both values plateau in range from around $n \sim 200$ for 2 threads and to around $n \sim 500$ for 10–12 threads and then show slight variations. For a fixed number of harmonics, increasing the number of threads generally leads to an increase in Speedup and a decrease in Efficiency.

While a few measurements show speedups below 1, the median and average speedups of approximately 3.5 demonstrate that the parallel algorithm generally achieves at least a threefold reduction in runtime. Maximum value of the speedup is 8.37 and achieved on $n = 2000$ and 16 threads. This run achieved an efficiency of 52% while average and median values are around 55.5% and roughly 30% of all tests resulted in efficiency over 75%.

Another important characteristic is the overhead

$$O_K = K \cdot T_K - T_1, \tag{20}$$

which quantifies the total additional CPU time wasted due to parallelization overheads (thread management, synchronization, load imbalance, and cache effects) relative to the

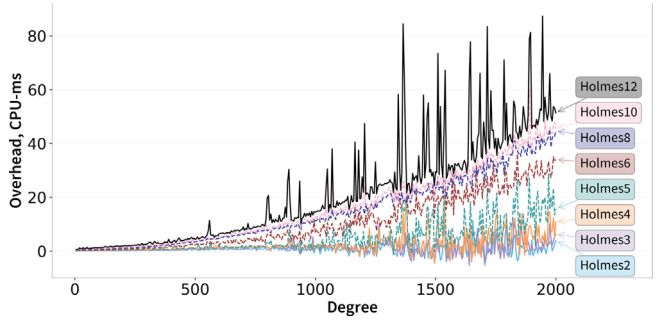


Fig. 7. Overhead of parallel Holmes algorithm over maximum degree for threads from 2 to 12 for processor P6T12.

ideal case of perfect scaling. Unlike efficiency, which is normalized by the number of threads, the overhead provides an absolute measure of these losses in core-seconds. Fig. 7 shows the overhead as a function of maximum degree for different thread counts on the P6T12 processor. For configurations using 2–5 threads, the overhead remains relatively stable across the range of degrees. In contrast, for higher thread counts it increases noticeably faster with degree, with the largest overhead observed for 12 threads. This behavior is likely attributable to hyper-threading, as the P6T12 processor has only 6 physical cores. So the additional virtual threads introduced by hyper-threading appear to amplify contention for shared resources. The particularly high overhead at 12 threads also underscores the greater resource waste in this configuration, consistent with the lower parallel efficiency observed for the same case in Fig. 6. By examining speedup, efficiency, and overhead together, practitioners can select the optimal degree of parallelism and expansion order that minimizes both wall-clock time and resource waste for a given computational budget.

To illustrate the practical relevance of the parallel Holmes algorithm, we assessed its performance within a realistic orbit propagation scenario using the EGM2008 gravity model truncated at degree and order 2000. Rather than employing a full-featured numerical integrator, we developed a dedicated simulator that isolates the computational cost of the spherical harmonic evaluation. This simulator, available in the same GitHub repository as the parallel Holmes implementation, mimics a fourth-order Runge–Kutta integrator with a fixed 1-s integration step. Each step requires four evaluations of the gravitational acceleration via the harmonic expansion. Between these evaluations, a placeholder workload function introduces artificial computational overhead (primarily matrix multiplications) to emulate the typical CPU time spent on coordinate transformations, auxiliary force model computations, and other integration-related tasks in a propagator. This approach is not a complete high-fidelity orbit integration with a full force model, but a controlled benchmark designed to highlight the relative contribution of high-degree harmonic computations in realistic propagation workloads. The simulation of a single day of orbit propagation was executed on a P4T8 processor. On a single thread, the total execution time was approximately 58681 s. With 8 threads, the time decreased to 28545 s, yielding a speedup of approximately 2.05. The observed speedup is lower than the values reported in Table 2. This difference arises because the workload function introduces extra computational time of approximately 60 ms per iteration. Over 345,600 steps it adds 20,736 s to both single-threaded and multi-threaded algorithms. Therefore the speedup decreases from the ideal 4.31 (pure gravity evaluation) to the observed 2.05. These results demonstrate that the parallel Holmes algorithm delivers substantial wall-clock time savings in scenarios representative of precise orbit determination and gravity field validation, making routine use of ultra-high-degree harmonic expansions practical.

4.2. Accuracy test

The primary advantage of accelerated algorithms, including the parallel Holmes method, is realized only if numerical accuracy and stability are preserved relative to established serial implementations.

We performed a direct comparison between serial and parallel implementations to show that parallelization itself does not introduce numerical differences. Gravitational accelerations were evaluated at 1000 points with randomly generated coordinates for maximum orders and degrees 10, 100, 500 and 1000. Fig. 8 shows relative deviation between the gravitational accelerations computed by serial and parallel implementations. The differences are $\leq 5.7 \cdot 10^{-14}$ which confirms the accuracy of parallel implementation compared to the serial one.

Another important aspect is the agreement level between Holmes algorithm and other algorithms. We evaluated gravitational acceleration over the same set of 1000 random independent points using the aforementioned Belikov's method as well as Fukushima's extended-range arithmetic method (Fukushima, 2011). Fig. 9 shows relative deviations between the Holmes algorithm and the other methods. The Holmes algorithm shows good agreement with reference methods with deviations not exceeding $1.02 \cdot 10^{-12}$. These comparisons confirm the accuracy of Holmes method itself and its parallel implementation.

To verify the accuracy of the parallel implementation in orbit propagation, we conducted controlled numerical integrations of a representative satellite orbit, comparing the gravitational accelerations and resulting position trajectories computed by the single-threaded Holmes and parallel Holmes algorithms.

We used Everhart's numerical integration propagation technique (Everhart, 1974) and accounted for Earth's rotation in frame transformations between celestial and terrestrial reference frames. Transformation matrices included precession, nutation and pole motion and were computed

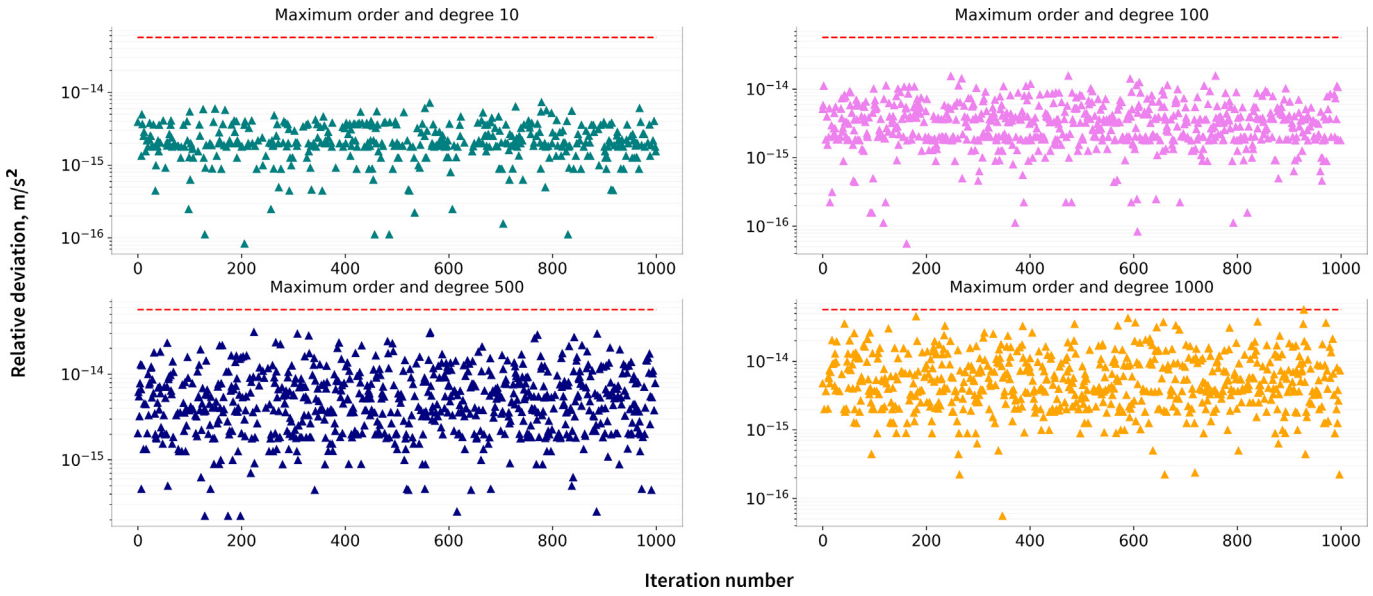


Fig. 8. Comparison of serial and parallel implementations of Holmes algorithm. Red dashed line shows maximum relative deviation which does not exceed $5.7 \cdot 10^{-14}$ m/s^2.

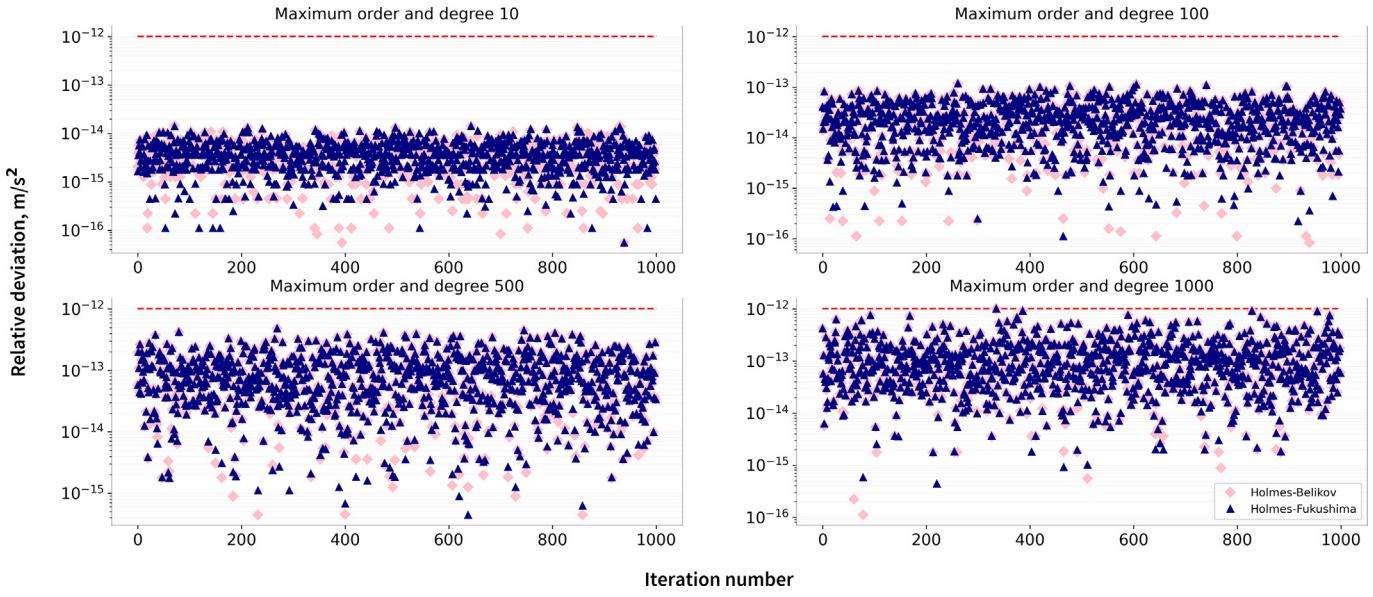


Fig. 9. Comparison of serial Holmes, Belikov and Fukushima algorithms. Red dashed line shows maximum relative deviation which does not exceed $1.02 \cdot 10^{-12}$ m/s^2.

using open SOFA library (see https://iausofa.org). The sole purpose was to isolate and compare the relative accumulation of numerical errors in the repeated evaluation of the gravitational acceleration from the errors arising purely from multi-threading technique.

The test case was designed to approximate circular low-Earth orbit at an altitude of approximately 500 km above the Earth's mean radius and inclination of 0° (equatorial). The initial velocity was directed tangentially to the radius-vector with magnitude equal to the local circular velocity $\sqrt{GM/r_0}$. Propagation was performed over 6000 s for the spherical harmonic expansion truncated at maximum degree and order 10, 20, 50, 100 and 500.

To test the accuracy of the multi-threaded algorithm we compare its deviations from some reference orbit with deviations of a single-threaded algorithm. We adopt as a reference orbit an equatorial orbit with 500 km altitude evaluated by single-threaded algorithm using model up to degree and order 2000. Fig. 10 shows absolute deviations of multi-threaded Holmes algorithm from the reference orbit, for truncations at 10, 20, 50, 100 and 500 degrees. The absolute deviation decreases with increasing order

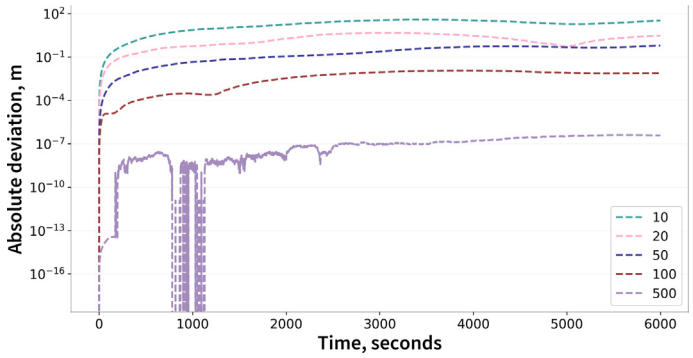


Fig. 10. Absolute position deviations from the full-model reference orbit (n = m = 2000, single-threaded) when using truncated models (n = m = 10, 20, 50, 100, 500) computed with the multi-threaded Holmes algorithm.

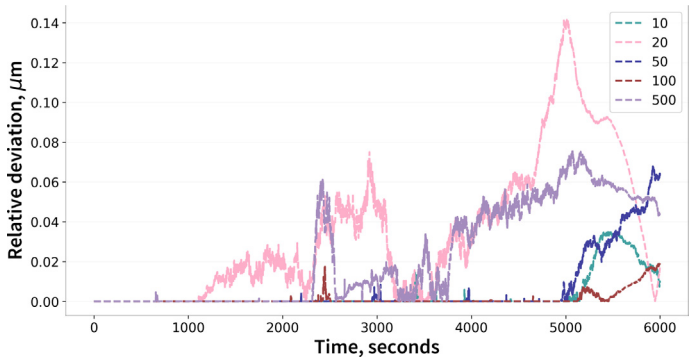


Fig. 11. Single-threaded and multi-threaded Holmes algorithms comparison on the equatorial orbit.

(and degree) of the model with results for 100 and 500 harmonics dropping to a millimeter-scale. This result correlates with the one from (Kuga and Carrara, 2013).

The same divergences were computed for the single-threaded Holmes algorithm for the same array of truncations. Fig. 11 shows absolute differences between single-threaded divergences and multi-threaded ones. The differences between the single- and multi-threaded versions remain below one micrometre for all tested truncations – three orders of magnitude smaller than the deviations observed at the 100-harmonics truncation.

An attempt to perform the same comparison using the near-polar orbit (inclination of 89.99°) revealed that neither single-threaded nor multi-threaded Holmes algorithms were able to compute accelerations on a reference near-polar orbit. The reason behind this inability is the factor of $\frac{1}{\sin(\theta)}$ in the $\frac{dP}{d\theta}$ (see (Holmes and Featherstone, 2002), Eq. 35). While evaluated for 2000 harmonics it overflows (as for 89.99° inclination $\theta = 0.01°$) causing all computations result in NaN values.

Overall, these results indicate that the multi-threaded Holmes algorithm maintains excellent numerical accuracy while providing significant computational advantages, making it well suited and preferable for high-resolution geopotential evaluations on non-polar orbits in celestial mechanics and astrodynamics applications.

5. Conclusion

This study has presented a multi-threaded parallel implementation of Holmes' algorithm for the efficient pointwise evaluation of high-degree spherical harmonic expansions of Earth's gravitational field. Parallelization on standard multi-core CPUs is achieved by distributing the outer order-wise summation (m-loop) across independent threads, with subsequent combination of partial sums in a master thread. The proposed method gives speedups, with the maximum value of 8.37 achieved for a 2000 × 2000 decomposition using 16 threads and a corresponding parallel efficiency of 52%. The approach scales well on contemporary hardware, enabling routine use of ultra-high-degree gravity models in precise long-term orbit propagation. Numerical tests confirm that the parallel version preserves the accuracy and stability of the serial Holmes algorithm. All implementations discussed in this paper—including the single-threaded Belikov and Cunningham methods, the serial and parallel Holmes algorithms, and benchmarks on random and sequential evaluations – were developed in C++20 and are publicly available at github.com/timurezy/3AKB. These results underscore the relevance of optimized recursive techniques in modern astrodynamics and provide an open-source tool for the community.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the use of the computing resources provided by the Astro Space Center.

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